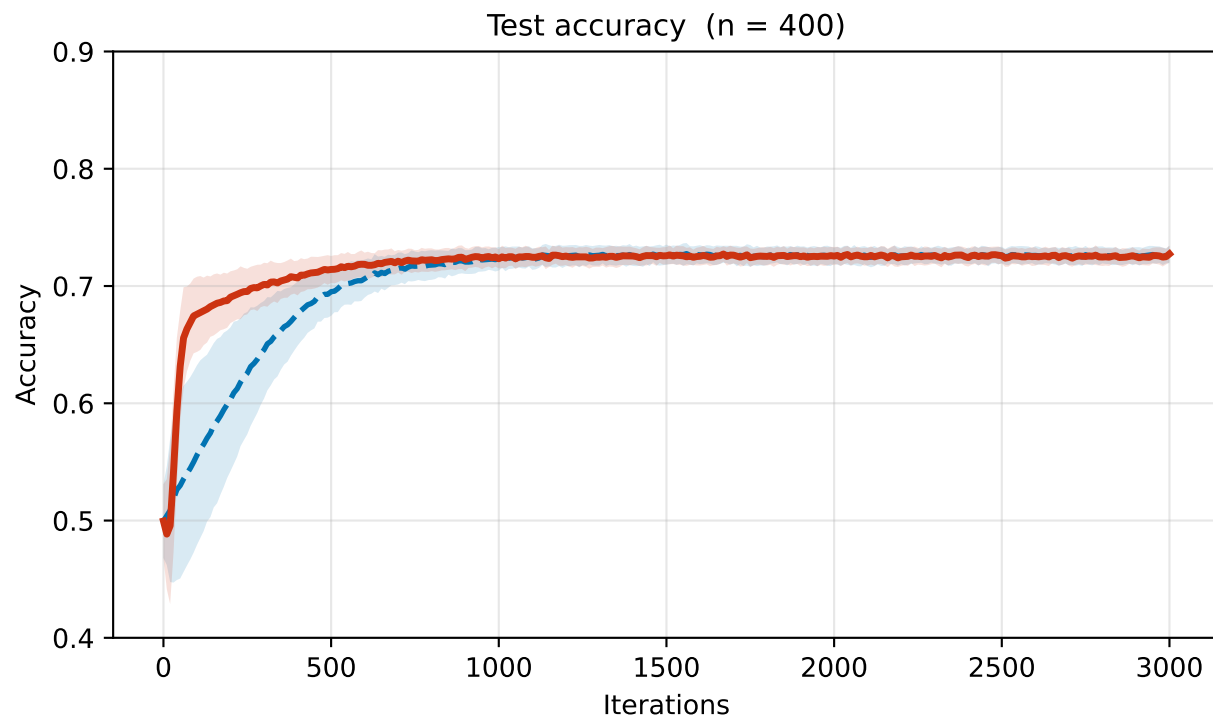
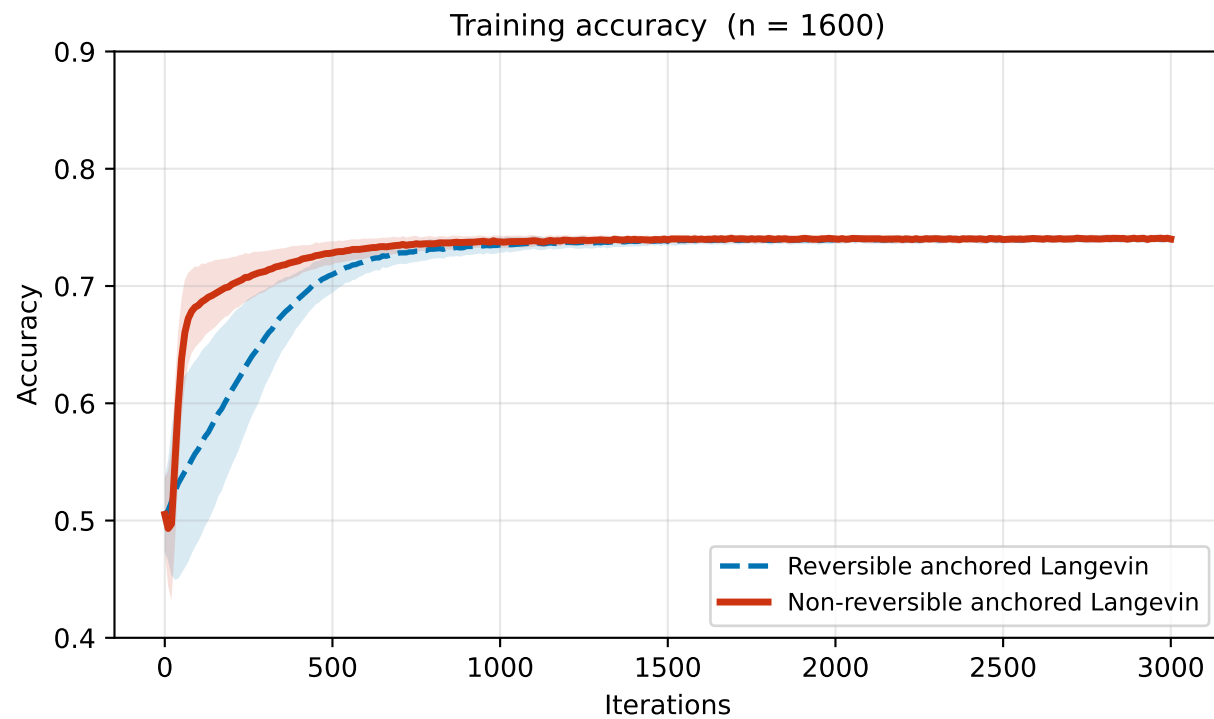


Non-reversible beats reversible anchored Langevin — unit ball



d = 9 (intercept + 8 slopes); constraint: unit ball; $U = f + g$ with $g = 2 \sum |w_j|$ (non-differentiable), anchor delta = 0.02; EXACT gradient; eta = 2.0e-06; s = 16; R = 100; alpha = 0 vs 1.

Design: unstandardised covariates -- in each slope triple the covariance is $4 q_1 q_1' + 256 q_2 q_2' + 4 q_3 q_3'$ with $q_1 = (1, 1, 1)/\sqrt{3}$ (the rotation axis), $q_2 = (0, 1, -1)/\sqrt{2}$ (fast, no signal), $q_3 = (2, -1, -1)/\sqrt{6}$ (slow, in the rotated plane, carries signal); beta_triple = (0.5, 0.125, 0.125).

Lines: across-replicate mean; bands: mean +/- 1 sample sd of single-iterate accuracy (repeat-run variability, not confidence intervals). Shared initialisation and noise per replicate. y-axis fixed to (0.4, 0.9) -- NOT windowed on the plateau.